Strong convergence for a repeated GUE interaction on three tensor legs

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13 October 2026

Abstract

Let X_N be a normalized Gaussian unitary ensemble matrix of size N^2. We place the same realization on two adjacent pairs of tensor factors, as $X_N \otimes I_N$ and $I_N \otimes X_N$ on $(\mathbb{C}^N)^{\otimes 3}$. For every fixed complex noncommutative polynomial, its operator norm converges in probability to the norm of that polynomial in two free variance-one semicircular elements. The argument computes the covariance of the dependent pair and applies existing Gaussian spectral comparison and operator-valued free convergence results. A direct Gaussian variance estimate supplies the trace convergence needed for the norm lower bound.

1 Introduction

A local interaction may be used repeatedly in a quantum spin system. The resulting operators act on different tensor factors, while their random coefficients come from one matrix. Collins and Yuan [3, Section 1.1 and Section 5(2)] describe this repeated-interaction question and study tensor embeddings with independent local GUE matrices. The more general tensor models in Chen, Garza-Vargas and van Handel [2, Section 9.4, Theorem 9.8] also use independent local matrices. Here we retain the common Gaussian coefficients and consider two overlapping ordered embeddings.

For each positive integer N, let X_N be a normalized Hermitian GUE matrix on $\mathbb{C}^N \otimes \mathbb{C}^N$. Thus its diagonal entries have variance N^{-2}, and its upper-triangular complex entries satisfy $\mathbb{E}|(X_N)_{ab}|^2 = N^{-2}$. Define

$$A_N = X_N \otimes I_N, \qquad B_N = I_N \otimes X_N \quad \text{on } (\mathbb{C}^N)^{\otimes 3}. \tag{1}$$

Both expressions use exactly the same X_N, with the displayed tensor factor order. Let s_1, s_2 be centered free semicircular elements of variance one in their reduced free-product C^*-algebra, equipped with its faithful tracial representation.

Theorem 1.1. *For every fixed polynomial $P \in \mathbb{C}\langle x, y\rangle$,*

$$\|P(A_N, B_N)\| \xrightarrow[N\to\infty]{\mathbb{P}} \|P(s_1, s_2)\|.$$

All norms are operator norms, and the norm on the right is the norm in the reduced free-product C^*-algebra.

This proves the three-site, two-interaction case of the repeated-matrix question. The conclusion includes non-self-adjoint polynomials and does not impose a degree restriction on a fixed polynomial. Larger families of repeated interactions are outside the present result.

The general comparison and convergence results are due to Bandeira, Boedihardjo and van Handel [1], Gao and Kunnawalkam Elayavalli [4], and Jekel, Lee, Nelson and Pi [6]. Our application consists of the covariance calculation for the shared realization and the passage through complete linear matrix pencils. The joint entry covariance has norm $2/N$, while every fixed covariance expression formed from the identity tends to its scalar free value. These facts permit comparison with the free pair. We keep the constant matrix coefficient inside the Gaussian comparison, and use the deterministic linearization of Haagerup and Thorbjørnsen [5]. Expected trace convergence and a variance bound then give the reverse norm inequality.

We write Tr for unnormalized trace and $\mathrm{tr}_k = k^{-1}\,\mathrm{Tr}$. Hilbert–Schmidt norms are unnormalized: $\|R\|_{\mathrm{HS}}^2 = \mathrm{Tr}(R^*R)$. The proof estimates are made for $N \geq 2$; a finite initial dimension does not affect the limit.

2 Covariance of the shared realization

Put $m = N^2$ and $n = N^3$. Choose a Hermitian orthonormal basis $(H_\alpha)_{\alpha=1}^{N^4}$ of $\mathrm{M}_m(\mathbb{C})$ for $\mathrm{Tr}(H_\alpha H_\beta) = \delta_{\alpha\beta}$. For the given matrix, the coordinates $g_\alpha = N\,\mathrm{Tr}(H_\alpha X_N)$ are independent standard real Gaussians. Hence

$$X_N = \frac{1}{N}\sum_\alpha g_\alpha H_\alpha, \qquad C_{1\alpha} = \frac{H_\alpha \otimes I_N}{N}, \qquad C_{2\alpha} = \frac{I_N \otimes H_\alpha}{N}. \tag{2}$$

With $Y_{1,N} = A_N$ and $Y_{2,N} = B_N$, both coordinates are given by

$$Y_{i,N} = \sum_\alpha g_\alpha C_{i\alpha}. \tag{3}$$

In particular, the scalar index is shared in the two sums.

The covariance maps on $\mathrm{M}_n(\mathbb{C})$ are

$$\eta_{ij,N}(R) = \mathbb{E}[Y_{i,N} R Y_{j,N}] = \sum_\alpha C_{i\alpha} R C_{j\alpha}.$$

The block map $R \mapsto (\eta_{ij,N}(R))_{i,j=1}^2$ is completely positive: stack $C_{1\alpha}$ and $C_{2\alpha}$ to form its Kraus operators. Since these matrices are Hermitian, cyclicity of trace gives

$$\mathrm{tr}_n(\eta_{ij,N}(R)T) = \mathrm{tr}_n(R\eta_{ji,N}(T)). \tag{4}$$

All component maps are normal in finite dimension.

Lemma 2.1. *For $i, j \in \{1, 2\}$,*

$$\sum_\alpha C_{i\alpha}^2 = I_n, \qquad \|\eta_{ij,N}(R)\| \leq \|R\|, \tag{5}$$

$$\eta_{11,N}(I_n) = \eta_{22,N}(I_n) = I_n, \qquad \eta_{12,N}(I_n) = N^{-2}U_N, \quad \eta_{21,N}(I_n) = N^{-2}U_N^*, \tag{6}$$

$$\mathrm{Tr}(C_{i\alpha}C_{i\beta}) = N^{-1}\delta_{\alpha\beta}. \tag{7}$$

Here U_N is the cyclic permutation unitary $U_N(e_a \otimes e_b \otimes e_c) = e_b \otimes e_c \otimes e_a$. The joint entry-covariance operator, in the unnormalized Hilbert–Schmidt convention, has norm $2/N$.

Proof. Basis completeness and Hermitian symmetry give

$$\sum_\alpha (H_\alpha)_{ab}(H_\alpha)_{cd} = \delta_{ad}\delta_{bc}, \qquad \sum_\alpha H_\alpha^2 = mI_m.$$

The second identity proves the first part of (5). The coefficient row $(C_{i\alpha})_\alpha$ has norm one, as does the corresponding adjoint column. Factoring the map as row, diagonal copies of R, and column proves the map bound.

The entry covariance of the original matrix is $\mathbb{E}[(X_N)_{ab}(X_N)_{cd}] = N^{-2}\delta_{ad}\delta_{bc}$. Using the ordered embeddings, the row (i,j,k) and column (l,p,q) entry of $\mathbb{E}[A_N B_N]$ is

$$N^{-2}\delta_{ip}\delta_{jq}\delta_{kl}.$$

This is $N^{-2}U_N$; taking adjoints gives the other cross term. The diagonal identity values follow from the first part of the lemma.

Equation (7) follows from $N^{-2}\operatorname{Tr}(H_\alpha H_\beta)\operatorname{Tr}(I_N) = N^{-1}\delta_{\alpha\beta}$.

Let V_N have columns $(\operatorname{vec} C_{1\alpha}, \operatorname{vec} C_{2\alpha})$ in the direct sum of two Hilbert–Schmidt spaces. The joint covariance operator is $T_N = V_N V_N^*$. Thus

$$V_N^* V_N = \frac{2}{N} I_{N^4}, \qquad \|T_N\| = \frac{2}{N}. \tag{8}$$

This is also the Choi operator in the convention of [6, Definition 1.5 and Remark 1.6]. □

For the limiting scalar base algebra $\mathbb{C}$, put $\eta_{ij}(z) = \delta_{ij} z$. A deterministic covariance expression means a fixed finite expression formed from scalars and the identity using sums, products, adjoints and the maps $\eta_{ij,N}$. Its coefficients and expression do not depend on N.

Lemma 2.2. *For each such expression f, there is a constant K_f such that*

$$\|f(\eta_N, I_n) - f(\eta, 1)I_n\| \le K_f N^{-2}, \qquad N \ge 2. \tag{9}$$

Consequently both its normalized trace and its norm converge to their scalar values.

Proof. Induct over the finite expression. Scalars and the identity give the base cases. Sums and adjoints preserve the estimate. For products use $R_N S_N - cdI = (R_N - cI)S_N + c(S_N - dI)$; the preceding estimates bound S_N uniformly. For a map application, write $R_N = cI_n + E_N$. Lemma 2.1 gives

$$\|\eta_{ij,N}(R_N) - \delta_{ij} c I_n\| \le \|E_N\| + |c| N^{-2}\mathbf{1}_{\{i\neq j\}}.$$

This proves the induction. The trace conclusion follows from $|\operatorname{tr}_n(E_N)| \le \|E_N\|$. □

The cross map in Lemma 2.1 is only small at the identity. The argument uses its uniform norm bound for general inputs; it does not require that its full operator norm tend to zero.

3 The spectral upper bound

Let (h_α) be a free variance-one semicircular family in a faithful tracial von Neumann algebra, and define

$$S_{i,N} = \sum_\alpha C_{i\alpha} \otimes h_\alpha. \tag{10}$$

The same h_α occurs in both components. By [6, Proposition 1.8], this is an M_n-valued semicircular family with covariance η_N.

Lemma 3.1. *For every fixed complex polynomial P,*

$$\|P(S_{1,N}, S_{2,N})\| \longrightarrow \|P(s_1, s_2)\|.$$

Proof. Use [4, Corollary 1.4 and Examples 2.5(3)], in the tracial form explained in [6, Theorem 4.3]. The finite base algebras are M_n, the base tuples are I_n, and the limiting base is $\mathbb{C}$ generated by 1. The covariance is block completely positive and trace-symmetric by (4); its diagonal value at 1 is bounded by one. Lemma 2.2 proves strong convergence of all deterministic covariance expressions, including normalized traces.

The norm tests in [4, Examples 2.5(3)] can also be read directly: for fixed p, q in the base variable,

$$\|p(I_n) - \eta_{ij,N}(q(I_n))\| \longrightarrow |p(1) - \delta_{ij} q(1)|.$$

The generation condition is imposed on the limiting base, not on the finite algebras. The faithful product trace on $\mathrm{M}_n \otimes W^*(h_\alpha)$ provides the tracial realization of (10). The chosen limit is the target free pair, with scalar covariance δ_{ij}. □

We need this convergence for matrix pencils as well as scalar polynomials. The following argument also covers continuous spectra.

Lemma 3.2. *Fix $d \geq 1$ and self-adjoint $a_0, a_1, a_2 \in \mathrm{M}_d(\mathbb{C})$, and set*

$$Q_{N,\mathrm{free}} = a_0 \otimes 1 + a_1 \otimes S_{1,N} + a_2 \otimes S_{2,N}, \qquad Q = a_0 \otimes 1 + a_1 \otimes s_1 + a_2 \otimes s_2.$$

There is a sequence $\delta_N \to 0$ such that

$$\mathrm{Spec}(Q_{N,\mathrm{free}}) \subseteq \mathrm{Spec}(Q) + [-\delta_N, \delta_N].$$

Proof. Put $\mathcal{A}_N = C^*(1, S_{1,N}, S_{2,N})$ and $\mathcal{A} = C^*(1, s_1, s_2)$. The norm in $\prod_N \mathcal{A}_N / \bigoplus_N \mathcal{A}_N$ is the norm limsup, where the direct sum denotes norm-null sequences. Indeed norm-null corrections preserve the limsup, and cancelling finite initial segments gives the reverse inequality. Lemma 3.1 therefore makes

$$P(s_1, s_2) \longmapsto [(P(S_{1,N}, S_{2,N}))_N]$$

a well-defined isometry on the polynomial algebra. It extends to a unital $*$-embedding of $\mathcal{A}$ into this quotient. Apply the map entrywise in M_d. Matrix boundedness and norm-nullness are equivalent to the corresponding properties of their finitely many entries.

If the claimed spectral containment failed, uniform boundedness would give a subsequence of spectral points λ_N tending to a real $\lambda \notin \mathrm{Spec}(Q)$. The inverse of $\lambda - Q$ maps to an inverse of the matrix sequence class. Choose a bounded representative R_N. Both inverse residuals tend to zero. For all sufficiently large N, their norms are less than $1/2$, so the Neumann series gives an actual two-sided inverse of $\lambda - Q_{N,\mathrm{free}}$, with norm at most $2 \sup_N \|R_N\|$.

Inversion sends λ_N to the spectral point $(\lambda - \lambda_N)^{-1}$ of that inverse. Its norm must tend to infinity, a contradiction. If $\lambda = \lambda_N$, invertibility already fails. This uses spectral points, without an eigenspace assumption. The sequence-quotient argument is the deterministic mechanism in [5, pp. 758–760]. □

Now consider the Gaussian pencil

$$Q_N = a_0 \otimes I_n + a_1 \otimes A_N + a_2 \otimes B_N = a_0 \otimes I_n + \sum_\alpha g_\alpha L_{\alpha,N}, \quad L_{\alpha,N} = a_1 \otimes C_{1\alpha} + a_2 \otimes C_{2\alpha}. \tag{11}$$

Its free replacement is precisely $Q_{N,\mathrm{free}}$. For a self-adjoint Gaussian pencil $Y = L_0 + \sum_\alpha g_\alpha L_\alpha$ of size D, the parameters in [1, Section 2.1] are

$$\sigma(Y)^2 = \left\| \sum_\alpha L_\alpha^2 \right\|, \qquad v(Y)^2 = \sup_{\mathrm{Tr}(R^* R) \leq 1} \sum_\alpha |\,\mathrm{Tr}(L_\alpha R)|^2.$$

The source's parameter σ_* satisfies $\sigma_* \le v$. Consequently [1, Theorem 2.1] gives, for all $t \ge 0$,

$$\mathbb{P}\Big\{\mathrm{Spec}(Y) \subseteq \mathrm{Spec}(Y_{\mathrm{free}}) + C\big[v(Y)^{1/2}\sigma(Y)^{1/2}(\log D)^{3/4} + v(Y)t\big][-1,1]\Big\} \ge 1 - e^{-t^2}. \tag{12}$$

Here $Y_{\mathrm{free}} = L_0 \otimes 1 + \sum_\alpha L_\alpha \otimes h_\alpha$. The deterministic shift is unchanged, and is not included in either noise parameter.

For (11), let $M_a = \|a_1\| + \|a_2\|$ and $K_a = 2(\|a_1\|_{\mathrm{HS}}^2 + \|a_2\|_{\mathrm{HS}}^2)$. Lemma 2.1 gives

$$\sum_\alpha L_{\alpha,N}^2 = a_1^2 \otimes I_n + a_2^2 \otimes I_n + N^{-2}(a_1 a_2 \otimes U_N + a_2 a_1 \otimes U_N^*),$$

so $\sigma(Q_N) \le M_a$. Bessel's inequality using (7) gives, for each component,

$$\sum_\alpha |\operatorname{Tr}((a_i \otimes C_{i\alpha})R)|^2 \le N^{-1}\,\|a_i\|_{\mathrm{HS}}^2\,\|R\|_{\mathrm{HS}}^2\,.$$

Applying two-term Cauchy–Schwarz to the shared coefficient sum yields $v(Q_N)^2 \le K_a/N$.

Choose $t_N = (\log(dN^3))^{3/2}$ in (12). Then, with probability at least $1 - e^{-(\log(dN^3))^3}$,

$$\mathrm{Spec}(Q_N) \subseteq \mathrm{Spec}(Q_{N,\mathrm{free}}) + [-r_N, r_N], \tag{13}$$

where

$$r_N \le C\left[K_a^{1/4} M_a^{1/2} N^{-1/4} (\log(dN^3))^{3/4} + K_a^{1/2} N^{-1/2} (\log(dN^3))^{3/2}\right] \longrightarrow 0.$$

The failure probabilities are summable. Borel–Cantelli and Lemma 3.2 give, for each fixed pencil and each $\varepsilon > 0$,

$$\mathrm{Spec}(Q_N) \subseteq \mathrm{Spec}(Q) + [-\varepsilon, \varepsilon] \quad \text{eventually, almost surely.} \tag{14}$$

The comparison was applied to the whole pencil, including a_0; no addition of separate spectral containments is used.

Proposition 3.3. *On a probability space carrying the GUE sequence,*

$$\limsup_{N\to\infty} \|P(A_N, B_N)\| \le \|P(s_1, s_2)\| \quad \textit{almost surely for every } P \in \mathbb{C}\langle x, y\rangle.$$

Proof. Intersect the events in (14) for all positive integer d and all self-adjoint coefficient matrices with entries in $\mathbb{Q} + i\mathbb{Q}$. This is a countable intersection. Coordinate pencils give almost-sure boundedness of A_N, B_N.

Fix such an outcome and form $\mathcal{C} = \prod_N \mathrm{M}_{N^3} / \bigoplus_N \mathrm{M}_{N^3}$, with coordinate classes y_1, y_2. For any rational self-adjoint pencil, a point outside $\mathrm{Spec}(Q)$ is at a fixed positive distance from every sufficiently late finite Gaussian spectrum. The late resolvents are uniformly bounded; setting the finitely many early resolvents to zero gives an inverse in the matrix quotient. Thus its spectrum is contained in $\mathrm{Spec}(Q)$. The deterministic [5, Theorem 2.4] gives a unital $*$-homomorphism from $C^*(1, s_1, s_2)$ onto $C^*(1, y_1, y_2)$, sending s_i to y_i. Its contractivity and the quotient norm-limsup formula give the conclusion for all complex polynomials. This is also the criterion stated in [1, Theorem 7.7]. □

4 Trace convergence and the norm lower bound

The expected trace theorem [6, Lemma 3.9] applies to our joint Gaussian tuple, with deterministic base data I_n. Lemma 2.1 supplies trace symmetry and bounded component maps; (8) gives Choi norms tending to zero also for every coordinate restriction. Lemma 2.2 gives every deterministic covariance-expression trace limit. The identity base is independent of the Gaussian tuple and has uniformly bounded moments. Hence, for each word w,

$$\mathbb{E}\,\mathrm{tr}_n\, w(A_N, B_N) \longrightarrow \tau w(s_1, s_2). \tag{15}$$

Here τ is the faithful trace of the limiting free pair.

We first bound the operator-norm moments needed for a direct Poincaré estimate. In (12), use the coordinate pencil A_N. Its parameters are $\sigma(A_N) = 1$ and $v(A_N) = N^{-1/2}$ by (5) and (7). The free coordinate norms are uniformly bounded by Lemma 3.1, and $N^{-1/4}(\log(N^3))^{3/4}$ is bounded. Since $\|A_N\| = \|B_N\| = \|X_N\|$, constants $M, C > 0$ independent of $N \geq 2$ satisfy

$$\mathbb{P}\{\|X_N\| > M + Ct\} \leq e^{-t^2}, \qquad t \geq 0. \tag{16}$$

Partition into $[0, M]$ and $(M + Ck, M + C(k+1)]$, $k \geq 0$. For each fixed positive integer r,

$$\sup_N \mathbb{E}\,\|X_N\|^r \leq M^r + \sum_{k\geq 0} (M + C(k+1))^r e^{-k^2} < \infty.$$

Lemma 4.1. *For every word w of length $\ell \geq 1$ there is a constant C_ℓ independent of N such that*

$$\mathrm{Var}(\mathrm{tr}_n\, w(A_N, B_N)) \leq C_\ell N^{-4}.$$

Consequently the normalized traces of all words converge almost surely to their traces in the free pair.

Proof. Write $F_N = \mathrm{tr}_n(Y_{i_1,N} \cdots Y_{i_\ell,N})$. Differentiate with respect to the actual shared scalar g_α in (3). Cyclicity gives

$$\partial_\alpha F_N = \frac{1}{n} \sum_{t=1}^{\ell} \mathrm{Tr}(C_{i_t,\alpha} W_t),$$

where W_t is the cyclic product of the remaining $\ell - 1$ factors. Using Cauchy–Schwarz over the positions and Bessel's inequality for (7),

$$\sum_\alpha |\partial_\alpha F_N|^2 \leq \frac{\ell}{Nn^2} \sum_t \|W_t\|_{\mathrm{HS}}^2 \leq \frac{\ell^2}{Nn} \|X_N\|^{2\ell-2}.$$

The Gaussian Poincaré inequality [5, Proposition 4.1 and Corollary 4.2] applies to this complex polynomial and its square-integrable gradient. The uniform moments just proved and $n = N^3$ give the variance bound. Constant words have zero variance.

For each positive ε, the squared-deviation inequality gives

$$\mathbb{P}\{|F_N - \mathbb{E}F_N| > \varepsilon\} \leq C_\ell \varepsilon^{-2} N^{-4}.$$

This is summable, so Borel–Cantelli gives $F_N - \mathbb{E}F_N \to 0$ almost surely. Combine with (15) and intersect the countably many word events. Finite linear combinations then cover all complex polynomial traces. No independence between different N is used. □

Proof of Theorem 1.1. Place the GUE matrices on a common probability space. For a fixed P, put $Z_N = P(A_N, B_N)^* P(A_N, B_N)$ and $Z = P(s_1, s_2)^* P(s_1, s_2)$. For every fixed integer $q \geq 1$, Lemma 4.1 gives $\mathrm{tr}_n(Z_N^q) \to \tau(Z^q)$ almost surely. Since Z_N is positive,

$$\liminf_N \|P(A_N, B_N)\| \geq \tau(Z^q)^{1/(2q)}.$$

The limiting trace is faithful, so $\tau(Z^q)^{1/q} \to \|Z\|$. Indeed the upper bound follows from $Z \leq \|Z\| \, 1$, and a nonzero spectral neighborhood below $\|Z\|$ has positive trace, giving the reverse bound after taking roots. The case $Z = 0$ is immediate. This is the trace-power argument of [6, Lemma 4.1]. Taking $q \to \infty$ after the fixed-q limit proves

$$\liminf_N \|P(A_N, B_N)\| \geq \|P(s_1, s_2)\| \, .$$

Combine with Proposition 3.3 to obtain the norm limit almost surely on this coupling. It therefore holds in probability for the prescribed GUE marginals. In the displayed model, I_N is the N by N identity, both embeddings keep the ordered legs, and P has fixed complex coefficients independent of N. □

References

[1] Afonso S. Bandeira, March T. Boedihardjo and Ramon van Handel, *Matrix concentration inequalities and free probability*, Invent. Math. **234** (2023), 419–487. Version used: arXiv:2108.06312v2.

[2] Chi-Fang Chen, Jorge Garza-Vargas and Ramon van Handel, *A new approach to strong convergence II: The classical ensembles*, arXiv:2412.00593v3 (2026).

[3] Benoît Collins and Wangjun Yuan, *Strong convergence for tensor GUE random matrices*, arXiv:2407.09065v1 (2024).

[4] David Gao and Srivatsav Kunnawalkam Elayavalli, *Toeplitz exactness for strong convergence*, arXiv:2604.14106v3 (2026).

[5] Uffe Haagerup and Steen Thorbjørnsen, *A new application of random matrices:* $\mathrm{Ext}(C^*_{\mathrm{red}}(\mathbb{F}_2))$ *is not a group*, Ann. of Math. **162** (2005), 711–775. doi:10.4007/annals.2005.162.711.

[6] David Jekel, Yoonkyeong Lee, Brent Nelson and Jennifer Pi, *Strong convergence to operator-valued semicirculars*, arXiv:2506.19940v3 (2026).

AI use

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